Annual Energy Usage and CO2 Emissions

By Maximilian Bahar

Background

The human race continues to use more and more energy every single year. But can we keep doing this? This project has made me consider the implications of my actions. Calculating a number that reflects my actions and comparing it with national averages and classmates' values was eye-opening. This project has further convinced me how important it is to be conscious of and take responsibility for our own actions.

Assumptions

- 6 weeks in Indonesia, 42 weeks in Boston.
- 4 one-way flights from Indonesia to Boston.
- In Indonesia, I drive a car and live with my parents.
- In Boston, I take electric trains and live in an apartment.
- Gas, electricity, clothing, and food statistics are based on regional averages.
- Other forms of consumption besides food and clothing are omitted.

Conversions

Conversion Efficiencies	**Reasoning**
Gas to Electricity ~30% Coal to Electricity ~35% Nuclear to Electricity ~35% Renewables to Electricity ~50%	For electricity generation, inefficiencies come from fuel transportation, power plant efficiencies, and electricity transmission. I assume that renewables have an aggregate efficiency of ~50%.
Crude Oil to Gasoline ~95% Crude Oil to Aviation Fuel ~95%	Crude oil is efficient in conversion since most of its products are "useful" (EIA, 2019). The ~5% loss reflects oil transportation and gasoline/aviation fuel distribution.
Natural Gas to Residential Gas ~95% Natural Gas to LPG ~95%	Natural gas is minimally refined, so inefficiencies come from gas transportation and distribution.
Textile industry "efficiency" ~30%	Clothing production is a complex process that includes production of clothes, advertising, distribution, etc. So, I assume that the total efficiency is ~30%.

Carbon Factors

- 56g CO2-eq/MJ natural gas (VQ, 2015).
- 94g CO2-eq/MJ coal (VQ, 2015).
- 63g CO2-eq/MJ LPG (VQ, 2015).
- 12g CO2-eq/MJ extraction-to-refinement of crude oil (ICCT, 2010).
- 3100g CO2-eq/kg aviation fuel (Verifavia, n.d.).
- 120g CO2-eq/km for gasoline (ecoscore, 2019).
- Nuclear and renewables electricity generation doesn't produce carbon.

Flights Calculation Example

Values needed for primary energy consumption:

- Boeing 787-900 with 239 person capacity (Japan Airlines, n.d.).
- 16,600 km 4 times a year (Google Earth, 2019).
- 1.91 gal jet fuel/km (Airways, 2014).
- 45 MJ/kg crude oil (The Physics Factbook, 2002).
- 3.27 kg/gal crude oil (Reference, n.d.).
- ~95% conversion efficiency, crude oil to aviation fuel.

To find MJ/year/capita:
= 45 MJ/kg * 3.27 kg/gal * 1/0.95 efficiency * 1.91 gal/km * 16,600km * 4 times/year * 1/239 people
= <u>82,194 MJ/year/capita</u>

Values needed for carbon emissions:

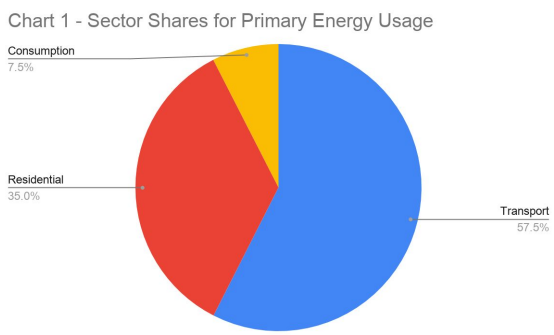
- 2 sources of emissions: extraction-to-refinement and combustion.
- 12g CO2-eq/MJ extraction-to-refinement of crude oil (ICCT, 2010).
- 3100g CO2-eq/kg aviation fuel (Verifavia, n.d.).
- 0.8 kg/liter aviation fuel (Verifavia, n.d.).

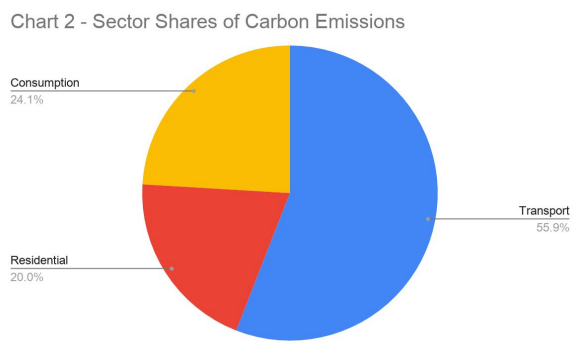
To find kg CO2-eq/year/capita:
= [(12g CO2-eq/MJ * 82,194 MJ/year/capita) + (3100g CO2-eq/kg * 0.8 kg/liter * 4.55 liter/gal * 1.91 gal/km * 16,600km * 4 times * 1/239 people)] * 1/1000 g/kg
= <u>6,758kg CO2-eq/year/capita</u>

Discussion of Results

Table 1 - Primary Energy and Carbon Emissions per Element

Sector	Element	Primary Energy (MJ)	Carbon Emissions (kg)
Transport	Flights	82,194	6,974
	Electric Train	2,153	97
	Private Car	479	38
Residential	Electricity	32,824	1,546
	Gas	18,768	1,000
Consumption	Food	10,800	3,000
	Clothing	322	59
MY TOTAL		**147,540**	**12,713**





- Transportation is my largest share of energy and carbon, probably due to frequent long distance flights paired with planes' low fuel efficiency and high carbon intensity.

- Food's share of carbon is larger than its share of energy. Possibly because food carbon emissions account for both energy-based emissions, like factories, and non-energy emissions, like methane from cows.

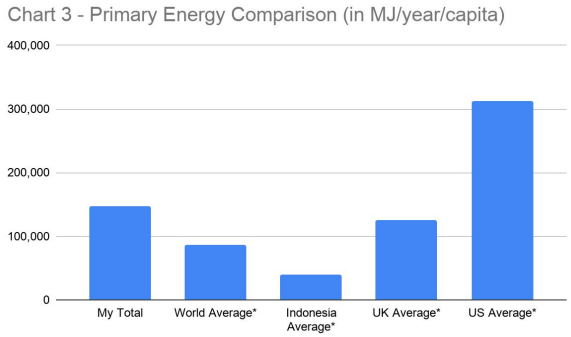
- Clothing consumption has a small share of both values. This could be caused by low energy 'costs' per product due to the large scale of the clothing industry.

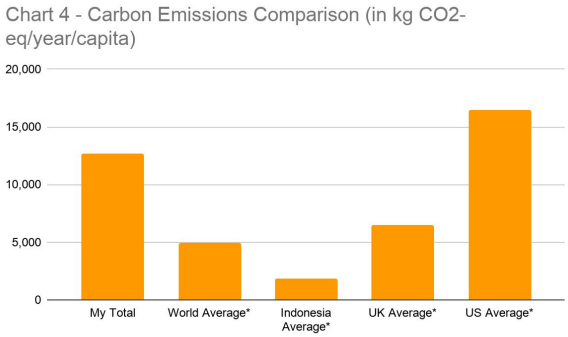
- Electricity's share of carbon is smaller than its share of energy. Possibly because of my incorrect assumption that nuclear and renewables don't produce carbon.

Global Comparisons

Table 2 - Global Comparisons for Primary Energy and Carbon Emissions

Element	Primary Energy (MJ)	Carbon Emissions (kg)
My Total	147,540	12,713
World Average*	86,445	4,981
Indonesia Average*	39,735	1,819
UK Average*	124,920	6,497
US Average*	313,200	16,503





- Developing countries like Indonesia have limited technology literacy, industrial activity, and, therefore, lower energy consumption. Which could explain why the Indonesian and World averages are lower than mine.

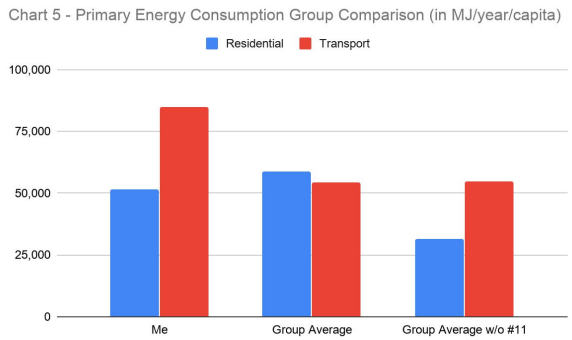
- Although my energy consumption is similar to the UK average, my carbon emissions are closer to the US average. It could be that the average person doesn't fly as far and often as I do. And since aircraft has high carbon intensity, I emit more carbon than the average person.

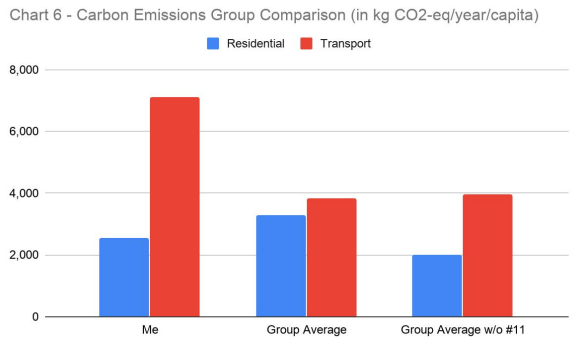
- Because national averages include per capita shares of industrial activity, the numbers suggests my energy consumption is underestimated.

Group Comparisons

Table 3 - Group Comparisons for Primary Energy and Carbon Emissions

Element	Primary Energy (MJ)	Carbon Emissions (kg)
Me	147,540	12,713
Group Average	130,087	8,509
Group Average w/o #11	108,605	7,805
*#11 determined to be an outlier		





*Our group has a high spread for both values, leading to biased averages

- Most likely due to my frequent flying, my share of energy consumption and carbon emissions for transportation is above the group average.

- My residential shares for both figures are slightly higher than the adjusted average. Possibly because a lot of our group calculated values for London, where energy efficiency is taken more seriously than the US.

Limitations

- Some products that are hard to quantify are omitted, i.e. soap, notebooks, etc. This may cause my energy consumption and carbon emissions to be underestimated.
- My share of industrial activity is also omitted, which may cause further biases for both values.
- Figures for food, clothing, electricity and gas are based on regional averages and won't reflect my actual values, since my habits will differ from the 'average'.
- My assumptions and estimations may not hold in real life, resulting in false results.

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